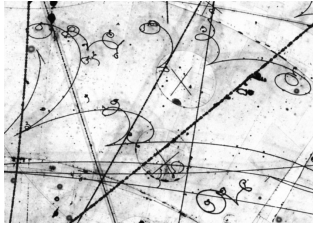


Alpha Lab Zero

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/Users/halpern/teaching/352/labs/2022/lab_alpha_zero2022.tex

Summary: The apparatus consists of an alpha particle source and a detector in a can. There is a pressure gauge, a vacuum pump to take the air out of the can, and a valve you can open to let air back in. The goal in general is to measure the full energy distribution of Alpha particles reaching the detector as you vary the amount of air in the can and to characterize the energy losses. An important part of this is to measure a Bragg curve of the differential energy loss per gram encountered.

However, in one day it might be enough to measure the rate of alpha particles when the can is at vacuum, and to measure an energy spectrum. In Lab Zero you measured the height of alpha particle pulses. The energy is proportional to this height, so what we want is a histogram of pulse heights.

The amplitude of the signals which the oscilloscope detects is proportional to the energy of the alpha particle. Another early exercise we recommend is to make a histogram of a few hundred energies with the chamber evacuated. How will you measure profile amplitude?

The tools you have are listed here. All are execute-permitted scripts in `/usr/local/bin`, which is in your path so you can run them by typing their name. For example, from the prompt in your directory, typing

```
ControlVacuum.py --help
```

runs ControlVacuum with the `help` flag set, and it returns a full menu.

ControlVacuum.py Run the `-help` option first.

ScopeTraceCapture.py grabs one Trace from Channel 1 on the scope, stores it in a png and displays it (should your X be working)

TriggerPulseDetection.py simply watches the pulses coming in on pin 5 from the trigger output of the scope. If that scope is set correctly to trigger on the alpha pulses, one can possibly estimate a pulse rate. Numbers are trigger time in microseconds.

ReadMM.py measures the DC voltage from the pressure sensor that is piped into channel 2 of the scope. 7.2V corresponds to vacuum, low voltage to an atmosphere of pressure. Work near the vacuum end.

ScopeRawQuery.py uses the measure facility of the oscilloscope to retrieve a value. You can change what you ask for. As above, edit this program to provide the data you actually want. Queries are faster than trace requests.

One important feature to notice: the commands to set a channel are different for retrieving a trace or a query. see the code. A particularly straightforward thing to do near vacuum is to measure the rate of alpha detections and whether they are random using **TriggerPulseDetection.py**.

You may want to alter these programs—copy them to your directory, change the name, and do whatever you want.

The Alpha computers are listed in Canvas.:

ssh in as **student** using the password **measure and learn** and create a folder of your own for programs and data.

Keep good notes of what you do.

The natural state of the apparatus is evacuated, with the valves closed and then the pump turned off. Please leave it like that when you are done for the day.